MODULE 16 · DOMAIN 3 - RESILIENT CLOUD SOLUTIONS

Disaster Recovery

RTO and RPO, the four strategies, AWS Backup and its vaults, Elastic Disaster Recovery, and proving recovery with Resilience Hub and Fault Injection Service.

DOMAIN 3 of 6 EXAM WEIGHT 15% MODULE 16 of 34

admin@sthithapragna.com

Two numbers pick the strategy: how much data you can **lose**, and how long you can be **down**. Then choose the **cheapest** design that meets both.

RPO and RTO

RECOVERY POINT OBJECTIVE

How much data you lose

- Measured backwards from the failure
- Set by how often data is copied
- Hourly backups: up to an hour lost
- Continuous replication: seconds

RECOVERY TIME OBJECTIVE

How long you are down

- Measured forwards from the failure
- Set by how much is already running
- Restore from backup: hours
- Traffic switch to a live copy: minutes

THE TIEBREAKER

RPO is about copying frequency. RTO is about readiness. They are improved by different spending.

The four disaster recovery strategies

STRATEGY	IN THE RECOVERY REGION	RPO / RTO	COST
Backup and restore	Backups and templates only	Hours / hours	Lowest
Pilot light	Data replicated; compute off or minimal	Seconds to minutes / tens of minutes	Low
Warm standby	A smaller, fully working copy, running	Seconds / minutes	Medium
Multi-site active-active	Full capacity, serving traffic	Near zero / near zero	Highest

ON THE EXAM

Pilot light: data live, servers ready to start. Warm standby: everything running, just smaller.

Making each strategy work

- **Infrastructure as code** rebuilds the recovery Region reliably. Without it, backup and restore takes days.
- AMIs, container images and secrets must already exist in the recovery Region.
- Pilot light and warm standby need a scaling step, then a traffic switch, as in module 14.
- Quotas in the recovery Region must allow full production capacity. Check them before the disaster.

> ON THE EXAM
>
> The most common DR failure is not data loss. It is discovering the recovery Region cannot run the application.

AWS Backup

- **Backup plans** define schedules, retention and lifecycle, and select resources by tag.
- Plans can **copy** backups to another Region and another account automatically.
- **Backup policies** in Organizations apply plans across every account.
- **Backup Audit Manager** reports whether resources meet your backup controls.

> ON THE EXAM
>
> Many services, many accounts, one policy, cross-Region copies: AWS Backup with an organisation backup policy.

Protecting the backups themselves

PROTECTION	WHAT IT STOPS	CAN IT BE UNDONE?
Standard vault with access policy	Casual deletion	Yes, by an administrator
Vault Lock, governance mode	Deletion without special permission	Yes, with that permission
Vault Lock, compliance mode	Deletion by anyone, even root	No, after the grace period
Logically air-gapped vault	Access from a compromised account	No: locked, and held outside your account

ON THE EXAM

Compliance mode is permanent once its grace period ends. Test with governance mode first.

Logically air-gapped vaults

- Backups live in an **AWS Backup service-owned account**, always under compliance-mode Vault Lock.
- Shared through **AWS RAM** to a recovery account, which restores directly, with no copy first.
- **Multi-party approval** (June 2025): an approval team can grant access even if your accounts are compromised.
- Since November 2025, a plan can back up to an air-gapped vault directly, as its primary target.

> ON THE EXAM
>
> Recover even if the production account, or the management account, is compromised: air-gapped vault plus multi-party approval.

Proving backups restore

- **Restore testing** runs scheduled restores in isolation, then deletes the test resources.
- Validation can run automatically, for example a Lambda function checking the restored data.
- Results show how long each restore took, which is evidence for your RTO.
- Backup Audit Manager proves backups exist. It does not prove they restore.

> ON THE EXAM
>
> Auditors want proof that backups actually restore, on a schedule: restore testing, not audit reports.

AWS Elastic Disaster Recovery

- Continuous **block-level replication** of servers, on-premises or EC2, into a low-cost staging area.
- RPO of seconds; recovery launches full instances in minutes.
- **Drills** launch recovery instances without interrupting replication.
- Point-in-time recovery helps roll back past ransomware; **failback** returns to the source.

> ON THE EXAM
>
> Servers that cannot be re-architected, seconds of RPO, minutes of RTO, low standby cost: Elastic Disaster Recovery.

AWS Resilience Hub

- Define an application from CloudFormation stacks, tags, AppRegistry or EKS clusters.
- A **resiliency policy** sets RTO and RPO targets for AZ, Region, infrastructure and application failures.
- Assessments report whether the design can meet them, with recommended alarms, runbooks and experiments.
- Run assessments from a pipeline through the API, and fail the release if the policy is breached.

> ON THE EXAM
>
> Check every release against RTO and RPO targets automatically: a Resilience Hub assessment in the pipeline.

AWS Fault Injection Service

- **Experiment templates** define actions, targets, an IAM role and stop conditions.
- Actions include stopping instances, adding latency, throttling APIs, and disrupting a whole AZ.
- **Stop conditions** are CloudWatch alarms that halt the experiment if harm goes too far.
- A **scenario library** offers ready-made experiments, such as an AZ power interruption.

> ON THE EXAM
>
> Resilience Hub assesses the design on paper. FIS tests the real system under real failure.

Service-level recovery building blocks

SERVICE	BUILT-IN RECOVERY	ACROSS REGIONS
RDS	Automated backups and point-in-time restore	Automated backup replication
Aurora MySQL	Backtrack: rewind in place, in minutes	Global Database
DynamoDB	Point-in-time recovery, up to 35 days	Global tables, or Backup copies
S3	Versioning; Object Lock for immutability	Replication
EBS and AMIs	Snapshots; Recycle Bin for deleted ones	Snapshot copy

ON THE EXAM

Undo a bad data change in minutes without a new cluster: Aurora Backtrack. Restore to a moment: point-in-time recovery.

Protecting against mistakes, not just disasters

- **Recycle Bin** retention rules keep deleted EBS snapshots and AMIs recoverable, for a set period.
- **S3 Object Lock** stops objects being deleted or overwritten, in governance or compliance mode.
- **Deletion protection** on databases and stacks stops a single call removing them.
- Most data loss is a mistake or a script, not a Region failure. Plan for both.

> ON THE EXAM
>
> A clean-up script deleted snapshots it should not have: Recycle Bin would have made them recoverable.

A Region failover, end to end

01
Detect
Alarms and health checks show the Region failing.

02
Decide
A person or a rule declares a disaster.

03
Promote data
Failover the database; confirm replication state.

04
Scale
Bring recovery capacity to production size.

05
Switch traffic
Route 53, Global Accelerator, or Region switch.

06
Verify
Synthetic tests confirm users are served.

ON THE EXAM

Region switch plans or Automation runbooks turn these steps into one tested, repeatable action.

Game days and drills

- Schedule recovery exercises, including the people and the runbooks, not just the technology.
- Measure the **actual** RTO and RPO achieved, and compare them with the targets.
- Use the built-in tools: Elastic Disaster Recovery drills, restore testing, Region switch plan evaluations.
- Record every gap found, fix it, and test again.

> ON THE EXAM
>
> The objective you state is a promise. The objective you measured in a drill is a fact.

> An internal reporting application can be unavailable for up to four hours after a Regional disaster, and can lose up to one hour of data.
>
> Its infrastructure is defined in CloudFormation. The company wants the lowest-cost DR design that meets these objectives.

Which strategy should the company use?

A Warm standby in a second Region

B Backup and restore: hourly AWS Backup copies to a second Region, with the stack redeployed from templates

C Pilot light, with a cross-Region read replica and stopped instances

D Multi-site active-active across two Regions

◆ PAUSE AND DECIDE

Which strategy should the company use?

A Warm standby

Recovers in minutes: far better than needed, at far higher cost.

B Backup and restore

Hourly copies meet the one-hour RPO; templates make a four-hour rebuild realistic. Cheapest.

C Pilot light

Works, but a continuously running replica costs more than the objectives require.

D Active-active

The most expensive option, for an internal tool that tolerates four hours down.

THE SIGNAL

Hours of RTO and an hour of RPO: backup and restore, made fast enough by infrastructure as code.

A company runs a legacy order system on EC2 instances whose state lives on EBS volumes. The application cannot be re-architected.

After a Regional failure it must recover within 20 minutes, losing no more than a few seconds of data, at the lowest standby cost.

Which solution meets these requirements?

A Daily AMIs copied to a second Region

B A full-size copy of the fleet running in a second Region

C Hourly EBS snapshots copied to a second Region with AWS Backup

D AWS Elastic Disaster Recovery replicating the instances to a second Region, with regular drills

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A Daily AMIs

Up to a day of data lost. Far outside a few seconds.

B Full-size running copy

Meets the times, but at the highest standby cost, and the data still needs syncing.

C Hourly snapshots

Up to an hour lost, and restoring volumes takes time.

D Elastic Disaster Recovery

Block-level replication gives seconds of RPO, minutes of RTO, and a cheap staging area.

THE SIGNAL

Legacy servers, seconds of RPO, minutes of RTO, low standby cost: Elastic Disaster Recovery.

> After a security review, a company fears that an attacker holding administrator access to its production account, or even its management account, could delete every backup.
>
> Critical backups must remain recoverable into a clean account even if those accounts are fully compromised.

Which solution meets this requirement?

A. Copy backups to a logically air-gapped vault, and associate a multi-party approval team from a separate recovery organisation

B. Enable Vault Lock in governance mode on the production account's vault

C. Copy backups to a second Region in the same account

D. Export backups to an S3 bucket with versioning enabled

◆ PAUSE AND DECIDE

Which solution meets this requirement?

A Air-gapped vault with approval team

Locked, held outside your accounts, and an independent team can grant a clean account access.

B Governance mode

An administrator with the bypass permission can still delete. The attacker is an administrator.

C Second Region, same account

The compromised account controls both copies.

D Versioned S3 bucket

Versions can be deleted by an administrator of the same account.

THE SIGNAL

Attacker owns the account, even management: the backup must live outside it, with an independent path to recover.

> Auditors now require monthly evidence that critical database backups can actually be restored, and how long each restore takes.
>
> The company already uses AWS Backup and wants the least operational overhead.

What should the DevOps engineer do?

A. Have an engineer restore a sample backup by hand each quarter

B. Enable Backup Audit Manager and send its reports to the auditors

C. Create an AWS Backup restore testing plan on a monthly schedule, with automatic validation

D. Write a scheduled Lambda function that restores snapshots and times them

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Quarterly by hand

Too infrequent for monthly evidence, and manual.

B Backup Audit Manager

Proves backups exist and follow policy. Not that they restore, nor how long it takes.

C Restore testing plan

Scheduled restores, automatic validation and clean-up, with durations recorded as evidence.

D Custom Lambda

Rebuilds restore testing by hand.

THE SIGNAL

Proof of restore, with timings, on a schedule: restore testing. Audit Manager proves existence, not recoverability.

> A company claims its application survives the loss of an Availability Zone, but has never tested it. It wants to prove this in a production-like environment.
>
> If customers begin to see errors during the test, it must stop automatically.

What should the DevOps engineer do?

A. Run an FIS experiment from the AZ power interruption scenario, with a CloudWatch alarm on error rate as a stop condition

B. Terminate the instances in one zone by hand and watch the dashboards

C. Run a Resilience Hub assessment of the application

D. Write a Lambda function that randomly stops instances every hour

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A FIS with a stop condition

A realistic zone failure, and the alarm halts it automatically if customers are affected.

B Terminate by hand

Only partly simulates a zone loss, and nothing stops it automatically.

C Resilience Hub

Assesses the design. It does not test the running system.

D Random-stop Lambda

Not a zone failure, no stop condition, and runs indefinitely.

THE SIGNAL

Prove it for real, stop automatically if harm appears: an FIS experiment with an alarm stop condition.

> A company has RTO and RPO targets for each of its applications. Architectural changes sometimes quietly break those targets, for example by removing a Multi-AZ setting.
>
> The company wants every release checked against its targets automatically, failing the pipeline if a change would breach them.

What should the DevOps engineer do?

A Deploy an AWS Config conformance pack for resilience best practices

B Review AWS Trusted Advisor's resilience checks before each release

C Define the application and a resiliency policy in Resilience Hub, and run an assessment from the pipeline, failing on a breach

D Run a Fault Injection Service experiment on every release in production

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Config conformance pack

Checks settings after deployment, and knows nothing of RTO or RPO targets.

B Trusted Advisor by hand

Manual, generic, and not tied to the company's targets.

C Resilience Hub in the pipeline

Assesses each release against its own RTO and RPO policy, and the pipeline can fail on a breach.

D FIS on every release

Risky and slow for every release, and it tests one failure, not the policy.

THE SIGNAL

Every release, against RTO and RPO targets, fail the pipeline: Resilience Hub. FIS proves; Resilience Hub assesses.

A nightly clean-up script deletes old EBS snapshots and AMIs. A bug made it delete snapshots still needed for recovery, and they could not be recovered.

The team wants deleted snapshots and AMIs to remain recoverable for 14 days, while the script keeps running, with the least effort.

What should the DevOps engineer do?

A. Copy every snapshot to another account each day with AWS Backup

B. Create Recycle Bin retention rules for EBS snapshots and AMIs with a 14-day retention period

C. Enable S3 Object Lock on the bucket that stores the snapshots

D. Attach an SCP that denies ec2:DeleteSnapshot

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A **Daily cross-account copies**

Protects data, but doubles storage and is far more effort than needed here.

B **Recycle Bin rules**

Deleted snapshots and AMIs stay recoverable for 14 days; the script is unchanged.

C **S3 Object Lock**

Snapshots are not objects in a bucket you control. There is nothing to lock.

D **SCP denying deletion**

Stops the script doing its legitimate job, and the storage grows forever.

THE SIGNAL

Deleted by mistake, recoverable for a window, cleanup still allowed: Recycle Bin.

> An order service runs on ECS with an Aurora database. After a Regional failure it must be serving again within 10 minutes, losing no more than a few seconds of data.
>
> The company wants the lowest cost that reliably meets these targets.

Which strategy should the company use?

A. Backup and restore with hourly cross-Region copies

B. Pilot light: Aurora Global Database secondary, with no ECS tasks running in the recovery Region

C. Active-active: full capacity in both Regions, serving traffic

D. Warm standby: Aurora Global Database secondary and a small ECS service running, scaled up on failover

◆ PAUSE AND DECIDE

Which strategy should the company use?

A Backup and restore

Loses up to an hour, and takes hours. Fails both targets.

B Pilot light

Data is fine, but starting and proving every service from nothing in 10 minutes is not reliable.

C Active-active

Meets the targets, at the highest cost. More than needed.

D Warm standby

Data within seconds, and a running, tested service that only needs scaling. Reliably inside 10 minutes.

THE SIGNAL

Seconds of RPO needs replicated data. Ten minutes of RTO needs something already running: warm standby.

Module 16, and Domain 3, in five lines

- RPO is copy frequency; RTO is readiness. Choose the cheapest strategy that meets both.
- Backup and restore, pilot light, warm standby, active-active: cost rises, time falls.
- AWS Backup copies across Regions and accounts; compliance lock and air-gapped vaults defeat attackers.
- Elastic Disaster Recovery for servers; restore testing proves backups work.
- Resilience Hub assesses designs against targets; FIS proves them under real failure.

ON THE EXAM

Domain 3 complete. Next, module 17 opens Domain 4: collecting logs and metrics.